

# Two-Gimbal PAT: An artistic installation combining mechanical and acoustic rotation of levitated content

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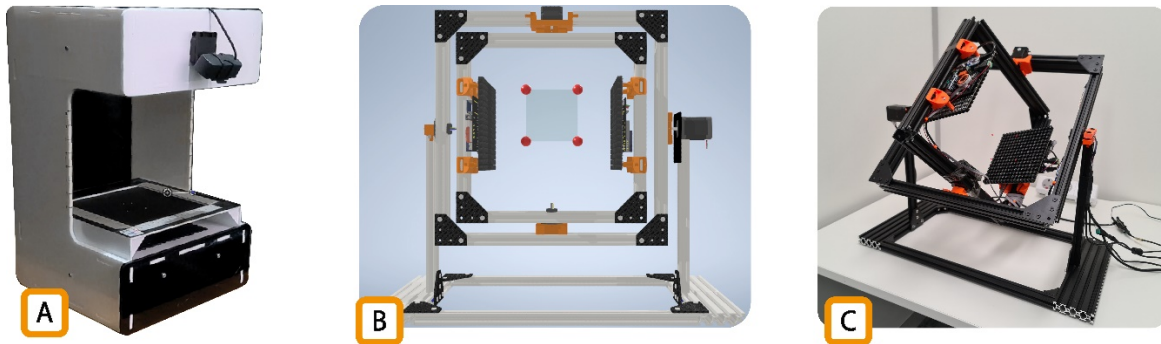


Figure 1: Acoustic levitation setups: a) static two-sided PAT, b) two-gimbal PAT 3D design and c) two-Gimbal PAT physical setup.

## ABSTRACT

Interactive applications using ultrasonic phased arrays of transducers (PATs) provide novel opportunities to create 3D displays. They show visual content using levitated particles such as polystyrene beads, thread or fabric, that dynamically change to render volumetric shapes in mid-air. Typically, this technology uses fixed arrays of transducers (e.g., two boards vertically or horizontally aligned) to display interactive content. That is, a fixed system of reference (SoR), defined by the levitation device, and dynamic mid-air content. This demo proposes a novel setup, featuring a dynamic-SoR for the device with the dynamic mid-air content. We do this by combining a PAT-based levitation device mounted onto a two-gimbal mechanism, resulting into a device with a dynamic coordinate system for volumetric animations. We describe our setup and demonstration, which aims to create an intriguing experience, where both the content and the device itself can freely move in 3D space, with the aim of challenging the user's expectations about the content displayed or even the principles enabling it.

## CCS CONCEPTS

• **Hardware** → Emerging technologies; Emerging interfaces.

## KEYWORDS

Phased Arrays of Transducers, acoustic levitation, artistic installations

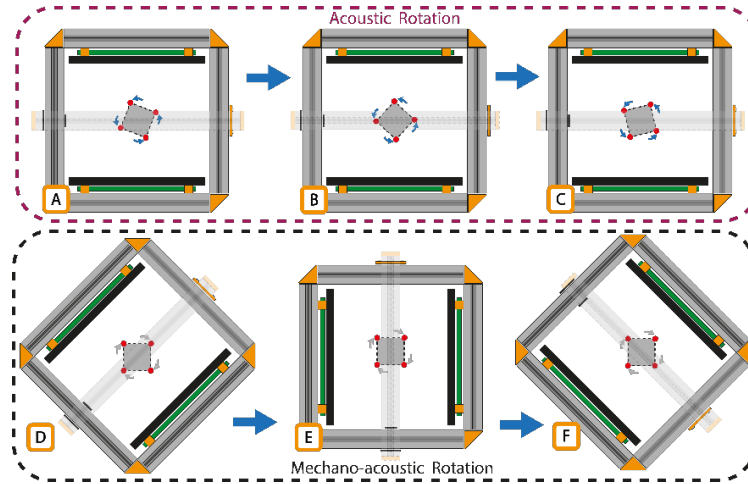
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## 1 INTRODUCTION

PATs allow acoustic levitation by using sound fields with covering forces at a point to trap particles in mid-air [1, 2]. The inherent properties of PATs have captured the attention not only from the research community on acoustics and HCI, but also from the general public due to the device capability to generate 3D volumetric visualizations without the need of additional stereo glasses. They also allow the user to experience simultaneous multimodal stimulus such as visual levitated content (using thread [3], fabric [10] or particles [6]), mid-air haptics without active gloves and parametric audio without special headphones. The device itself presents an open space that allows not only to display 3D content, but also to interact with it directly using bare hands, eyes and ears.

This technology has been used in different applications such as none-contact transportation [19], levitated food [21] and small insects [22] as well as chemical sample mixing [16, 23]. Particularly, in the area of volumetric displays, particle-based acoustic levitation is used in different ways to visualize information. For instance, in form of grids [12], 2D screens [17], dynamically updated charts



**Figure 2: Two-gimbal rotations: mechano-acoustic rotations starting from clockwise gimbal rotations (from a-e) and followed by acoustic rotations (from e to h).**

[14], enhancing on-display content [9] and visualizing trajectories [13] among others.

In all these examples, the PAT configurations always use a fixed coordinate system. For example, fixed setups using single [11], and multiple boards [5], vertical [4] and horizontal [12] alignments, ellipsoidal [8] and octagonal [18] arrangements or wearable approaches [7]. While those configurations allow dynamic visual rendering, the displayed content is limited to the static nature of the PAT's coordinate system. Therefore, exploring a dynamic setup to augment the rendered content has not been explored to date.

In this demo, we integrated two PAT boards inside a two-gimbal device mechanically actuated, to combine mechanical and acoustic motion on the levitated content. This allows us to propose two ways to manipulate the content displayed in mid-air (via acoustic or mechanical rotation). This demo paper describes this concept, details the example setup that enables it and provides a description of the kind of interaction that attendees to the conference will experience. For instance, the mechanical gimbals can rotate the device, while the levitated content remains fixed in space. Alternatively, gimbals can slowly come to a halt, while the levitated content starts to rotate. With this, we aim to create an intriguing experience, where users feel uncertain of the nature and role that each element plays (i.e., gimbals, PATs), while demonstrating the potential of PAT technologies to showcase meaningful mid-air content. Our demo aims to motivate and inspire other researchers and artists to explore novel applications for acoustic technologies using PATs and novel integrations between PATs and other technologies.

## 2 TWO-GIMBAL PAT CONCEPT

As shown in Figure 2, our two-gimbal mechanism enables both mechanical and acoustic rotations as well as the combination of both to generate novel types of animations together with the supported levitated content (e.g., single and multi-particle, fast moving content and LeviProps [10]). For instance, Figure 2(A-C) depicts the content rotation by using an acoustic-only approach from PATs. That is, the device remains fixed, and the acoustic field produces

animation. The device also supports the use of mechanical-only rotations in two axes. That is, the PATs are attached to the inner part of the device and the mechanical rotations of the gimbal will also produce a rotation in the content on the direction of the motion. However, by combining mechanical and acoustic rotations (in opposite directions) as shown in Figure 2 (D-F), we can generate the illusion of the content not being affected by the device rotations. In other words, the content remains static in the middle of the space no matter how the two-gimbal device rotates. This work not only exploits the use of multiple rotation types for levitated animations (i.e., mechanical rotations only, acoustical rotations only or the combination of both), but also turns the device itself on an active part of the animation, blinding with this the wall between the device and the content.

## 3 PROTOTYPE

Constructed is a dual-ring gimbal set supported through a central stand as shown in Figure 1 (C), the prototype permits rotation in two orthogonal axes with a central area of 16x16x16cm reserved for levitation (see Figure 1 (B-C)).

### 3.1 Mechanical Structure

Two gimbal rings are constructed from 20x40 mm v-slot extrusions, joined with right angle corner connectors and supported through a central stand. The central stand contains two flat 20x80 mm v-slot extrusions connected with 20x20 mm v-slot bracers, and two vertical 20x20 mm v-slot extrusions to support the two gimbals.

The central stand is connected to the outer gimbal through a NEMA17 stepper motor (1705HS200A), providing rotational torque to spin the outer gimbal. On the opposite side of the outer gimbal motor is a 3D printed part supporting a 12-wire slip ring (SRC022-12) that couples the central stand to the outer gimbal additionally carrying all electrical wiring.

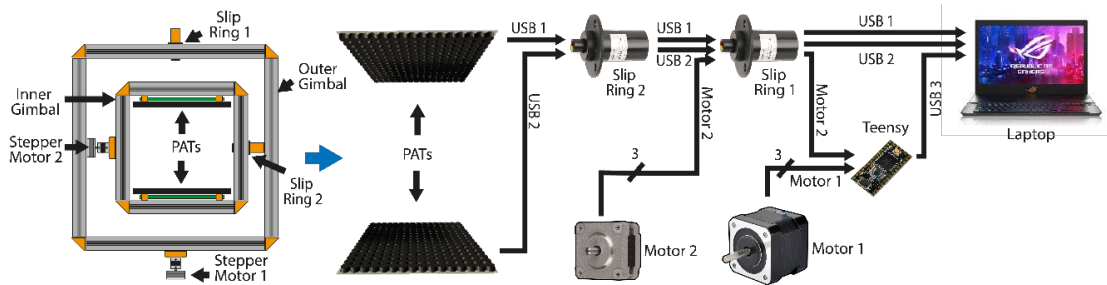


Figure 3: Hierarchical schematic of the prototype.

In a similar way, the outer gimbal is connected to the inner gimbal through a slip ring and stepper motor combination, this time using a smaller stepper motor (NEMA17 1701HS140A).

### 3.2 Electrical Design

Two USB buses connect a computer to the PATs alongside two 24 V DC rails from a power supply used to power the PATs. As shown in Figure 3, two differential pairs are sliced, twisted, shielded and terminated through both sets of slip rings connecting the computer to the PATs via USB to allow data transfer. Each terminal of the slip ring is rated 2 Amps and each PAT consumes up to 40Watts. Therefore, we use two 24VDC rails, passed through the slip rings to power the PATs.

The outer slip ring needs to also carry the signal for the inner stepper motor. Extra precautions need to be taken to shield the neighboring signal wires from interference, as the stepper motor signals tend to be noisy, high current tracks. The stepper motors themselves are powered from a third 24V power supply and controlled by two drivers which in turn are themselves controlled by microcontrollers, all of which is kept away from the central stand and gimbal structure, apart from some discreetly positioned wires.

### 3.3 Safety Considerations

It is worth mentioning that the PATs used in this demo, utilize the same transducers and voltages as commercial devices available for public exploitation (e.g., Ultraleap [20]). These devices have been cleared for public use and hence there are no associated risks to health to the visitors. Additionally, to avoid unsafe interactions due to the mechanical parts in motion we will not include the presentation of parametric sound or haptics rendering on this demo, so we avoid attendees approaching their hands or face to the device.

## 4 DEMONSTRATION OF TWO-GIMBAL PAT

This demo does not require the audience to use any additional instrumentation such as glasses to visualize the animated content. The demo is focused on inspiring researchers, artists, and the general audience to engage, interact and design applications with PAT devices and foster integrations with different technologies. Our demonstration is composed by three types of content (single fast-moving particle also called POV content [6], multi-particles shapes [15] and LeviProps or mini screens [10]) and four modes.

### 4.1 Content Types

There are multiple content types supported by PATs in the literature. However, for simplicity in this demo we are using:

*Single-particle POV*: A single polystyrene bead (2mm diameter) draws a shape at the center of the levitator device. This bead is slowly accelerated until it reaches the persistence of vision effect ( $\sim 10\text{Hz}$ ) [6] allowing a visualization of the full shape in the 3D space.

*Multi-particle shapes*: This content uses multiple particles to draw a geometrical shape in mid-air at a time (e.g., circle, triangle, square, pentagon, star, etc.).

*Mini-screen projection*: In this type of content, a LeviProp is used as a canvas to play projected content such as images and short videos while the LeviProp is rotating and moving around the 3D space.

### 4.2 Device Modes

This demo plays multiple animations in sequence, one after the other, combining one of the supported content types and one of the operational modes described below:

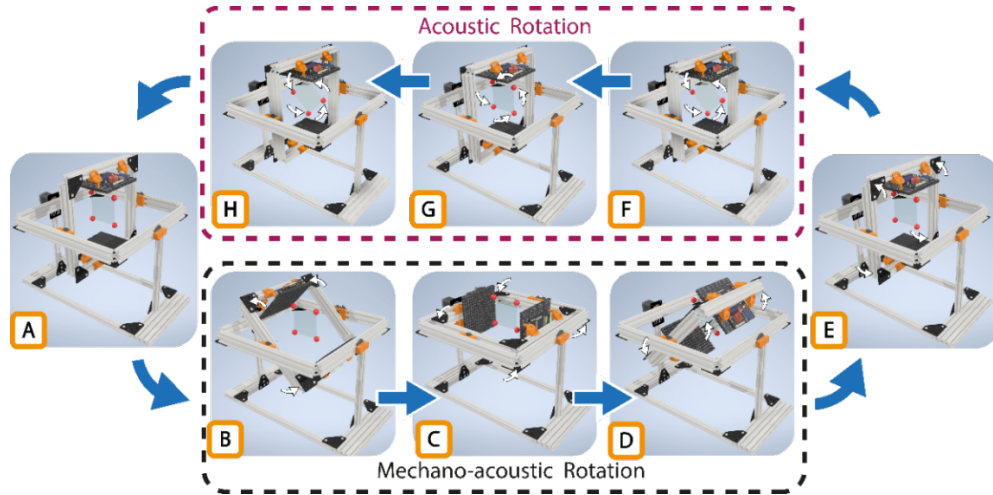
*“Fixed frame & Content spins”*: In this first mode, we use acoustic rotations to spin the content either clockwise or counterclockwise on one of the three axes at a time while the frame remains static.

*“Spinning frame & fixed content”*: The second mode uses mechanical rotations in one of the two axes at a time, while the content remains static with respect to PAT reference system. That is, as the levitator is attached to the inner gimbal, visually in this mode, the content will follow the device rotations.

*“Spinning frame & spatially fixed content”*: In this third mode, the device is spinning in any of the two supported axes, whilst the content is acoustically rotated always in the opposite direction of the device movement to generate the effect of it being static in the center of the two-gimbal device. In other words, the effect of spatially fixed content.

*“Fully independent”*: In this last mode, the rotations from the two-gimbal device and the ones acoustically generated are independent each other, generating content rotations and patterns difficult to predict for the audience.

Then, during the demo presentation, the audience will experience a combination of these four modes and the three types of content supported by the device, highlighting the integration of mechano-acoustic displacements we proposed on an installation



**Figure 4: Spinning frame & spatially fixed content: clockwise gimbal rotations (from a-e) while acoustically levitated content spins in opposite direction (static content effect), followed by acoustic rotations only (from e to h).**

where the device is merged with the animation, becoming an integral part of it. An example of one of the animations to experience in our demo, starts with a rotation of the inner gimbal (clockwise direction on the Z axis) while the outer gimbal remains static. At the same time, the levitated content (e.g., a LeviProp) starts spinning on the same axis and with the same movement ratio but to the opposite direction (counterclockwise) generating though the effect the content being static (“Spinning frame & spatially fixed content”) while the two-gimbal device is rotating (see Figure 4 A-E). After some full revolutions, the two-gimbal rotation stops, making visible the LeviProp rotation in counterclockwise direction as shown in Figure 4 (E-H).

## 5 DISCUSSION & CONCLUSION

In this demo, we introduced a non-fixed coordinate system approach for the levitated content, in which the center of the system of reference is changing according to the two-gimbal rotations. This integration between PATs and a two-gimbal mechanism allows us to showcase the use of mechano-acoustic rotations as an example of novel combination of technologies focused on encouraging the audience (researchers and artists) to get involved in the generations of a broader type of applications where the device is merged with the animation and the experience, not only in the research field, but also on content creation for artistic installations for a widespread diffusion of the PATs technology potential.

As a future work, we target to include an interactive control using mid-air gestures not only over the levitated content but also over the rotations of the two-gimbal mechanism. Additionally, we envision the use of this setup as a levitation testing kit where novice users can learn the effect of the physical forces acting on the levitated content such as inertial, friction and gravity forces, as well as learning the limits of PATs in terms of the acoustic field, trapping forces, particle acceleration and particle sizes.

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